

depending on whether ad hoc or infrastructure modes are used and, in the case of infrastructure mode, whether a wireless station or an AP is sending the frame.) Finally the WEP field indicates whether encryption is being used or not. (WEP is discussed in Chapter 8.)

6.3.4 Mobility in the Same IP Subnet

In order to increase the physical range of a wireless LAN, companies and universities will often deploy multiple BSSs within the same IP subnet. This naturally raises the issue of mobility among the BSSs—how do wireless stations seamlessly move from one BSS to another while maintaining ongoing TCP sessions? As we'll see in this subsection, mobility can be handled in a relatively straightforward manner when the BSSs are part of the subnet. When stations move between subnets, more sophisticated mobility management protocols will be needed, such as those we'll study in Sections 6.5 and 6.6.

Let's now look at a specific example of mobility between BSSs in the same subnet. Figure 6.15 shows two interconnected BSSs with a host, H1, moving from BSS1 to BSS2. Because in this example the interconnection device that connects the two BSSs is *not* a router, all of the stations in the two BSSs, including the APs, belong to the same IP subnet. Thus, when H1 moves from BSS1 to BSS2, it may keep its IP address and all of its ongoing TCP connections. If the interconnection device were a router, then H1 would have to obtain a new IP address in the subnet in which it was moving. This address change would disrupt (and eventually terminate) any on-going TCP connections at H1. In Section 6.6, we'll see how a network-layer mobility protocol, such as mobile IP, can be used to avoid this problem.

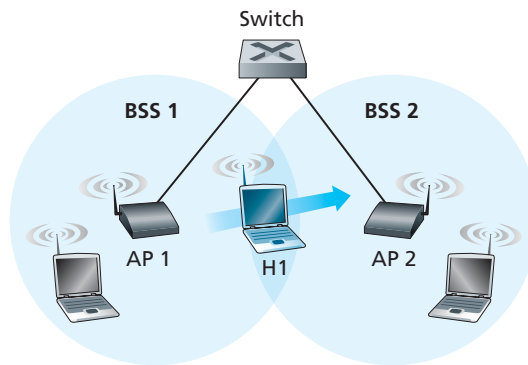


Figure 6.15 ♦ Mobility in the same subnet

But what specifically happens when H1 moves from BSS1 to BSS2? As H1 wanders away from AP1, H1 detects a weakening signal from AP1 and starts to scan for a stronger signal. H1 receives beacon frames from AP2 (which in many corporate and university settings will have the same SSID as AP1). H1 then disassociates with AP1 and associates with AP2, while keeping its IP address and maintaining its ongoing TCP sessions.

This addresses the handoff problem from the host and AP viewpoint. But what about the switch in Figure 6.15? How does it know that the host has moved from one AP to another? As you may recall from Chapter 5, switches are “self-learning” and automatically build their forwarding tables. This self-learning feature nicely handles occasional moves (for example, when an employee gets transferred from one department to another); however, switches were not designed to support highly mobile users who want to maintain TCP connections while moving between BSSs. To appreciate the problem here, recall that before the move, the switch has an entry in its forwarding table that pairs H1’s MAC address with the outgoing switch interface through which H1 can be reached. If H1 is initially in BSS1, then a datagram destined to H1 will be directed to H1 via AP1. Once H1 associates with BSS2, however, its frames should be directed to AP2. One solution (a bit of a hack, really) is for AP2 to send a broadcast Ethernet frame with H1’s source address to the switch just after the new association. When the switch receives the frame, it updates its forwarding table, allowing H1 to be reached via AP2. The 802.11f standards group is developing an inter-AP protocol to handle these and related issues.

6.3.5 Advanced Features in 802.11

We’ll wrap up our coverage of 802.11 with a short discussion of two advanced capabilities found in 802.11 networks. As we’ll see, these capabilities are *not* completely specified in the 802.11 standard, but rather are made possible by mechanisms specified in the standard. This allows different vendors to implement these capabilities using their own (proprietary) approaches, presumably giving them an edge over the competition.

802.11 Rate Adaptation

We saw earlier in Figure 6.3 that different modulation techniques (with the different transmission rates that they provide) are appropriate for different SNR scenarios. Consider for example a mobile 802.11 user who is initially 20 meters away from the base station, with a high signal-to-noise ratio. Given the high SNR, the user can communicate with the base station using a physical-layer modulation technique that provides high transmission rates while maintaining a low BER. This is one happy user! Suppose now that the user becomes mobile, walking away from